DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING			
Course Code: 24CS36L	*Course Title: Data Structures Lab*		
Total Credits: 1.5 L-0: T-0: P-45	*SDGs addressed: 4,9,11*	*Course Category: PCCL*	*Course Type: Laboratory*

Teaching Learning Process		*Total Contact Hours/semester*	*Assessment Weightage and Marks*			
	Lecture	-		CIE	SEE	Total
	Tutorial	-	*Weightage, %*	*40*	*60*	*100*
	Practical	45	*Maximum Marks*	*40*	*60*	*100*
	Self-learning and Term Work	-	*Minimum Marks*	*20*	*25*	*45**
	Total	45				

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

Course Prerequisite: C programming.

Course Objectives:

Sl. No.	Course Objectives
1	Gain knowledge in practical applications of data structures and to introduce various techniques for representation of the data in the real world.
2	Strengthening the ability of the students to identify and apply the suitable data structure for the given real-world problem.
3	Develop skills to design and analyze simple linear and nonlinear data structures.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Implement programs to illustrate the use of pointers to arrays, structures and functions.	L3	WK2, WK6
CO2	Implement various operations on Linear lists with Array and Linked list representations and analyze the performance of operations.	L3	WK2, WK4
CO3	Implement different data structures such as Stack, Queue, Tree and Graph using Array and Linked list representations with static or dynamic memory allocation and analyze the performance of operations, use stack for expression conversion and evaluation.	L3	WK2, WK4

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

Course Content / Syllabus:

Weeks	List of Programs	No. of Hours
1	Write programs for the following to illustrate the use of pointers. a) Create 3 pointers called P1, P2, P3 and an integer variable X where P1 points to P2, P2 points to P3 and P3 points to X. Demonstrate how to read and retrieve the value of X using P1, P2 and P3. b) Crate a pointer to an integer array, Read and display the contents using	03

	pointer and also display the contents in reverse order using pointer. c) Create a pointer to an integer array and pass this as parameter to a function called Reverse which reverses the array contents using pointer. Write different functions called Read (P) to read the array elements using pointer P and Display (P) to display the array contents using P before and after reverse. d) Create a pointer to two-dimensional array of integers. Write functions Read (P) and Display (P) to read and display the contents using pointer. e) Create two pointers called PS1 and PS2 to two strings, Read the contents using pointers and also write a function StrCat (PS1, PS2) which concatenates second string to the end of first string using pointers PS1 and PS2.	
2	Define a structure called Student with the members: Name, Reg_no, marks in 3 tests and average_ marks. Develop a menu driven program to perform the following by writing separate function for each operation: a) read information of N students b) display student's information c) to calculate the average of best two test marks of each student. Note: Allocate memory dynamically and illustrate the use of pointer to an array of structure.	03
3	Define a structure called Time containing 3 integer members (hour, minute, second). Develop a menu driven program to perform the following by writing separate function for each operation. a) Read (T): To read time b) Display (T):To display time c) update(T):To Update time d) Add (T1, T2) : Add two times. Update function increments the time by one second and returns the new time (if the increment results in 60 seconds, then the second member is set to zero and minute member is incremented by one. If the result is 60 minutes, the minute member is set to zero and the hour member is incremented by one. Finally, when the hour becomes 24, Time should be reset to zero. While adding two time variable, normalize the resultant time value as in the case of update function. Note: Illustrate the use of pointer to pass time variable to different functions.	03
4	Develop a menu driven program to implement the following operations on array storage representation with static and dynamic memory allocation. i)Insert by position ii) Delete by position iii)Insert by key iv) Delete by key v) Insert by order vi) Search by key vii) Search by position viii) Reverse the contents.	03
5	Develop a menu driven program to implement singly linked list with various operations such as i) Insertion and Deletion at front/rear ii) Insertion and Deletion at the specified position iii) Delete by Key iv) Search by key	03

	v) Create an ordered list vi) Reverse a list vii) Creating a copy of the list	
6	Develop a menu driven program to implement Circular singly linked list with various operations such as (Note: Use Header node to simplify he complex operations) i) Insertion and Deletion at front/rear ii) Insertion and Deletion at the specified position iii) Delete by Key iv) Search by key v) Create an ordered list vi) Reverse a list vii) Creating a copy of the list	03
7	Develop a menu driven program to implement Circular Double linked list with Header node to perform various operations such as i) Insertion and Deletion at front/rear ii) Insertion and Deletion at the specified position iii) Delete by Key iv) Search by key v) Create an ordered list vi) Reverse a list and vii) Creating a copy of the list	03
8	Develop a menu driven program to implement Stack with static and dynamic memory allocation mechanisms. (Represent Stack using structure)	03
9	a) Develop a menu driven program to convert infix expressions to postfix expression and evaluate the postfix expression. b) Develop a menu driven program to convert infix expression to prefix and evaluate the prefix expression.	03
10	a) Develop a menu driven program to implement ordinary Queue with static and dynamic memory allocation mechanisms. (Represent Queue using structure). b) Develop a menu driven program to implement Circular Queue with static and dynamic memory allocation mechanisms.(Represent Queue using structure)	03
11	a) Develop a menu driven program to implement Double Ended Queue with static and dynamic memory allocation mechanisms. (Represent Queue using structure) b) Develop a menu driven program to implement Priority Queue with static and dynamic memory allocation mechanisms (Represent Queue using structure)	03
12	a) Develop a menu driven program to implement binary search tree and traversal techniques. b) Develop a menu driven program to implement Graph traversal techniques.	03
13	**Revision**	03
14	**Revision**	03
15	**Laboratory Test**	03

Text Books:

Sl. No.	Author/s	Title	Publisher Details
1	Reema Thareja	Data Structures using C	2nd Edition, 2018, Oxford University Press.
2	Richard F.Gilberg and Behronz A. Forouzan	Data structures, A Pseudo code Approach with C	3rd Edition, Cengage learning, 2013

Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	Yedidyah, Augenstein, Tannenbaum	Data Structures Using C and C++	2nd Edition, PHI
2	E Balaguruswamy	Data Structures Using C	McGraw Hill Education, Paperback 2017
3	Ellis Horowitz and Sartaj Sahni	Fundamentals of Data Structures in C	2nd edition, Universities Press, 2014
4	Seymour Lipschutz	Data Structures Schaum's Outlines	Revised 1st Edition, McGraw Hill, 2014

Web Resources:

Sl. No.	Web Links
1	https://nptel.ac.in/courses/106102064/
2	https://nptel.ac.in/courses/106105085/

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	-	-	-	-	-	3	3	-	3	-	3
CO2	3	3	3	3	-	-	-	-	3	3	-	3	-	3
CO3	3	3	3	3	-	-	-	-	3	3	-	3	-	3

High – 3, Medium – 2, Low – 1